Hanhong Zhao

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Education

Massachusetts Institute of Technology
Undergraduate Study Sep 2024 - Now

IIIS, Tsinghua University
Pre-College Program Sep 2023 - Jun 2024

The High School Affiliated to Renmin University of China
Middle school Sep 2018 - Jun 2024

Awards

Gold Medal
53rd International Physics Olympiad 2023.7

Publications

One-step Latent-free Image Generation with Pixel Mean Flows
Yiyang Lu, Susie Lu*, Qiao Sun*, Hanhong Zhao*, Zhicheng Jiang, Xianbang Wang, Tianhong Li, Zhengyang Geng, Kaiming He*
https://arxiv.org/abs/2601.22158

Bidirectional Normalizing Flow: From Data to Noise and Back (CVPR 2026)
Yiyang Lu, Qiao Sun*, Xianbang Wang*, Zhicheng Jiang, Hanhong Zhao, Kaiming He*
https://arxiv.org/abs/2512.10953

Is Noise Conditioning Necessary for Denoising Generative Models? (ICML 2025)
Qiao Sun, Zhicheng Jiang*, Hanhong Zhao*, Kaiming He*
https://arxiv.org/abs/2502.13129

Experiences

Undergraduate Research in Computer Vision
Supervised by Professor Kaiming He Sep 2024 - Now

Collaborating with undergraduate students at MIT & IIIS.

- Developed and optimized large-scale JAX training pipelines on TPU clusters for group.
- Conducted research on generative models, with a major focus on diffusion models and their variants.
- Built text-to-image and multimodal generation/understanding systems, with experiments focusing on pixel-space generation and representation learning.

Course projects

Speeding Up Diffusion Models with One-step Generators
2024.12 `https://github.com/Hope7Happiness/6s978_project`

Project for the course *6.S978: Deep Generative Models* at MIT, completed with teammate Zhicheng Jiang.

- Proposed a new method to speed up the training of diffusion models by using one-step generators (such as VAEs).
- Reduces NFE by half while maintaining the sample quality on the MNIST dataset.

Knowledge Database Project
2024.5 `https://github.com/Hidden-Hyperparameter/LLM_project`

Project for the course *Introduction to Large Language Model Application* at IIIS, Tsinghua University, completed in a team of 3.

- Apply Large Language Models to answer user questions based on the knowledge in user-provided documents, and provide sources for the answers.
- Support multiple document type, including PDF, Word, images and videos.
- Build a tagging system to maintain a reasonable speed even with a large number of documents.

Skills

Programming Languages: Python, C++
Tools: PyTorch, JAX
Languages: English (Fluent), Chinese (Native)